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Concatenated Modular BK Channel Constructs Reveal Divergent Stoichiometry in Gating Control by LRRC26 ($\gamma 1$), Pore, and Selectivity Filter

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In this **important** contribution, Yan and colleagues describe a powerful and **compelling** strategy to generate concatamers of the BK channel and their fusion constructs with the auxiliary gamma subunits, which allows exploring contributions of individual subunits of the tetrameric channel to its gating and the study of heteromeric channel complexes of defined composition. Distinct examples are presented, which illustrate great diversity in the stoichiometric control of BK channel gating, depending on the site and nature of molecular perturbations. The molecular approaches could be extended to other membrane proteins whose N and C termini face opposite sides of the membrane.

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Abstract

Big-conductance, Ca^{2+} -activated K^+ (BK) channels consist of Ca^{2+} - and voltage-sensing, pore-forming α (BK α) subunits and regulatory auxiliary β or γ subunits. Concatenated subunit constructs are powerful tools for elucidating subunit stoichiometry in ion channel gating and regulation, allowing control over subunit arrangement, stoichiometry, and mutation. However, the additional S0 transmembrane segment in BK α places its N- and C-termini on opposite sides of the membrane, preventing tandem BK channel subunit construction by conventional methods. To investigate the atypical “all-or-none” modulatory function of γ subunits and the subunit stoichiometry of BK channel gating, we developed concatenated constructs containing 2 or 4 BK α subunits by splicing them into modular forms that can be co-expressed to form functional channels. These constructs retained voltage and Ca^{2+} gating properties similar to intact BK channels. By fusing the LRRC26 ($\gamma 1$) subunit to the N-terminus of tandem BK α constructs, we found that a single $\gamma 1$ subunit per α subunit tetramer is sufficient to fully modulate the channel. Furthermore, the L312A mutation in the deep pore region exhibited a stoichiometrically graded effect on voltage-gated BK channel activation. In contrast, a V288A mutation at the selectivity filter induced channel inactivation only when present in all four BK α subunits. Thus, by engineering concatenated BK α constructs, we identified three distinct stoichiometric modes of BK channel gating control by LRRC26, the pore, and the selectivity filter. This study offers new molecular tools and advances our understanding of subunit stoichiometry in BK channel gating and modulation.

Introduction

The big-conductance, calcium- and voltage-activated K^+ (BK) channel is a unique member of the potassium channel family, characterized by exceptionally large single-channel conductance and dual regulation by membrane voltage and intracellular free Ca^{2+} (1). The BK channel is a homotetramer composed of four identical pore-forming, Ca^{2+} - and voltage-sensing α (BK α) subunits (~130 kDa) and variable auxiliary subunits. BK channels exhibit prominent features in

molecular architecture (2, 3) and allosteric gating mechanisms (4, 5). In the transmembrane (TM) domains, BK channels differ from most voltage-gated K⁺ (Kv) channels by possessing an extra S0 TM helix, lacking domain swapping between the S1-S4 voltage-sensing domain (VSD) and the S5-S6 pore-gate domain (PGD), and exhibiting tight VSD-PGD packing via extensive S4-S5 interactions (2). The BK α subunit also contains a large cytosolic C-terminus composed of two tandem RCK domains (RCK1, RCK2) responsible for Ca²⁺ and Mg²⁺ sensing (5–11). The RCK domains from all BK α subunits assemble into a tetrameric two-layer gating ring that expands and shifts toward the membrane in response to Ca²⁺ bindings, leading to S6 movement and channel opening (8, 9).

BK channel function is regulated by auxiliary β and γ subunits and by LINGO1, conferring tissue-specific gating and pharmacological properties (12–18). The four γ subunits (γ 1– γ 4), also known as LRRC26, LRRC52, LRRC55, and LRRC38, are leucine-rich repeat (LRR)-containing membrane proteins (15, 16). The γ subunits facilitate BK channel activation by shifting the voltage dependence of channel activation in the hyperpolarizing direction by ~140 mV (γ 1), 100 mV (γ 2), 50 mV (γ 3), and 20 mV (γ 4), in terms of half-maximal activation voltage ($V_{1/2}$) in the absence of Ca²⁺. The γ 1 subunit likely modulates BK channels by enhancing the allosteric coupling between VSD activation and the pore opening (15). All γ subunits share a common topology, including an N-terminal signal peptide, an extracellular LRR domain, a single TM segment, and a short intracellular C-terminus (15, 16, 19). Their modulatory effects on BK channel voltage gating are mainly determined by the TM segments and C-terminal clusters of positively charged residues (20, 21), while the LRR domains regulate the γ subunits' expression and surface trafficking (19). Recent cryo-EM structures of BK α / γ 1 complexes show that the LRRC26's TM segment binds peripherally to the BK α VSD, involving S0, S2, S3, and pre-S1 helices, while the LRR domains tetramerize extracellularly without directly contacting BK α (22–24).

A fundamental question in ion channel gating and regulation is the structural and functional subunit stoichiometry. BK channel modulation by the γ subunits exhibits an atypical binary “all-or-none” phenotype: voltage-dependence ($V_{1/2}$) is either fully shifted or unchanged under limited γ 1 expression (19, 25). In tetrameric ion channels, auxiliary proteins typically follow 4-fold symmetry, producing graded modulation based on subunit stoichiometry relative to the pore-forming principal subunit, as observed with BK β subunits (26), KCNE subunits on KCNQ channels (27), and KChIP subunits on Kv4 channels (28). Consistent with previous reports detecting up to four γ 1 subunits per channel (29–31), Cryo-EM structures also show a symmetric presence of four γ 1 subunits in BK α / γ 1 complexes (22–24). However, single-channel recordings using a β 2- γ 1 chimeric subunit suggest that even a single γ 1 subunit is sufficient for full modulation (29). It is of note that the evidence remains inconclusive due to several limitations: the chimeric construct lacks the LRR domain, which may influence γ 1 function (19); the inferred subunit number is indirectly based on the β 2 N-terminal blockade effect; and the sample size in number of channels is limited by the single-channel recording method. Moreover, given the largely independent impact of the individual VSDs to BK channel gating (4) and the symmetric presence of γ 1 near all VSDs (22–24), an alternative concerted “all-subunits-required” model, involving extracellular LRR domain tetramerization, has been proposed (22). Therefore, direct biochemical determination of the functional stoichiometry of γ 1 in BK channel modulation is needed.

Concatenated subunit constructs with 2 or 4 channel subunits fused together in a C-to-N-terminal arrangement have been powerful tools for dissecting subunit stoichiometry and cooperativity in various voltage- and/or ligand-gated channels (32–38). However, the additional S0 segment in BK α prevents straightforward concatenation, as its N- and C-termini reside on opposite sides of the membrane. BK channels, owing to their unique biophysical properties, serve as a value model for studying allosteric gating mechanisms of multimodal ion channels (4). Yet, the unavailability of functional BK α concatemers has hindered precise stoichiometric investigations of channel gating and modulation at both intra- and inter-subunit levels.

To address this, we engineered modular BK α constructs that reassemble into functional concatenated channels with biophysical properties comparable to intact BK channels. Using these, we demonstrate that a single γ 1 subunit per BK α tetramer is sufficient to fully modulate the channel. Given the central role of the PGD in BK channel gating, we further applied this system to

mutational analyses of the deep pore and selectivity filter. We revealed distinct stoichiometric requirements for gating control by LRRC26, the pore, and the selectivity filter. This study provides new molecular tools and mechanistic insights into the stoichiometry of BK channel gating and regulation.

Results

Construction of functional BK channels with concatenated tandem repeats of the α subunits

We employed multiple strategies to generate concatenated BK α subunit constructs that enable expression of functional channels with biophysical properties that are comparable to intact (i.e., unsplit) BK channels and facile amplification and manipulation at the plasmid DNA level. Given the extracellular location of the N-terminus of BK α (Fig. 1A [↗](#)), which precludes direct C-to-N-terminal concatenation, we first attempted to split BK α into the N-terminal S0 part and the remaining major portion. This was based on the report that co-expression of these two parts can form functional channels ([39](#)). However, plasmids of tandem constructs lacking only S0 proved difficult to generate and use due to large plasmid size and instability during cloning. Since the C-terminal RCK2 domain can also be expressed as a module that forms functional channels when split and co-expressed with the rest of BK α ([40](#)), we thus further split BK α into a main part (residues 94-649), designated as α_M , and the rest by deletion of the main part, designated as $\alpha^{\Delta M}$ (Fig. 1A, B [↗](#)). The α_M module contains the major transmembrane region (S1 to S6) and the RCK1 domain. The $\alpha^{\Delta M}$ construct retains the N-terminal residues 1-93 including S0 and the C-terminal residues 652-1113 including the RCK2 domain.

To prevent homologous recombination within the concatenated tandem repeat constructs during molecular cloning and to facilitate site-directed mutations on specific subunits, we designed each BK α subunit in the repeats to have distinct DNA sequences. This was achieved by utilizing codon-optimized cDNA sequences for the 2nd, 3rd, and 4th repeats of the main part, which differ by ~25% in nucleotide sequence from the original 1st repeat and from each other. With these strategies, we generated a single unit and tandem constructs of double and quadruple repeats of α_M , named $\alpha_{M(\text{mono})}$ (α_M), $\alpha_{M(\text{dual})}$ (α_{MM}), and $\alpha_{M(\text{quad})}$ (α_{MMMM}), respectively (Fig. 1B [↗](#)). To evaluate the expression and stability of the concatenated tandem α_M constructs, we performed immunoblot analysis of the C-terminally V5-tagged constructs transfected in HEK293 cells. The result showed predominant protein bands of α_{MM} and α_{MMMM} at expected protein sizes recognized by the anti-V5 antibody (Fig. 1C [↗](#)), confirming the expression and lack of major degradation of the concatenated α_M constructs. The expression level of the $\alpha^{\Delta M}$ construct was affected by the α_M construct and co-immunoprecipitation confirmed the complex formation between the α_M and $\alpha^{\Delta M}$ constructs (Fig. 1 - figure supplement 1 [↗](#)).

By co-expression of the C-terminally GFP-tagged $\alpha^{\Delta M}$ ($\alpha^{\Delta M}$ -GFP) with single, double, or quadruple repeat constructs of α_M , we observed formation of functional BK channels that resemble the intact BK α channel in their voltage and Ca^{2+} -dependence of channel activation (Fig. 1D, E [↗](#)). The $V_{1/2}$ values of the BK channels formed by the single-unit α_M construct, when co-expressed with $\alpha^{\Delta M}$ -GFP, were 192 and 16 mV at virtual 0 and 10 μM Ca^{2+} , respectively. The tandem double repeat constructs, α_{MM} , when co-expressed with $\alpha^{\Delta M}$ -GFP, produced functional BK channels with $V_{1/2}$ values of 186 mV at virtual 0 Ca^{2+} , and 16 mV at 10 μM Ca^{2+} (Fig. 1E [↗](#); Table 1 [↗](#)). Furthermore, the channels formed by tandem quadruple repeat construct α_{MMMM} and $\alpha^{\Delta M}$ -GFP had $V_{1/2}$ values of 184 and 29 mV at virtual 0 and 10 μM Ca^{2+} , respectively (Fig. 1E [↗](#); Table 1 [↗](#)). These $V_{1/2}$ values of engineered BK channels formed by single, double, and quadruple repeat constructs of α_M are close to those of the intact BK α channels, which had $V_{1/2} = 172$ and 19 mV at virtual 0 and 10 μM Ca^{2+} , respectively (Table 1 [↗](#)). These results show that the voltage- and Ca^{2+} -gating properties are largely unaltered in the engineered BK channels formed by engineered concatenated BK α subunit constructs.

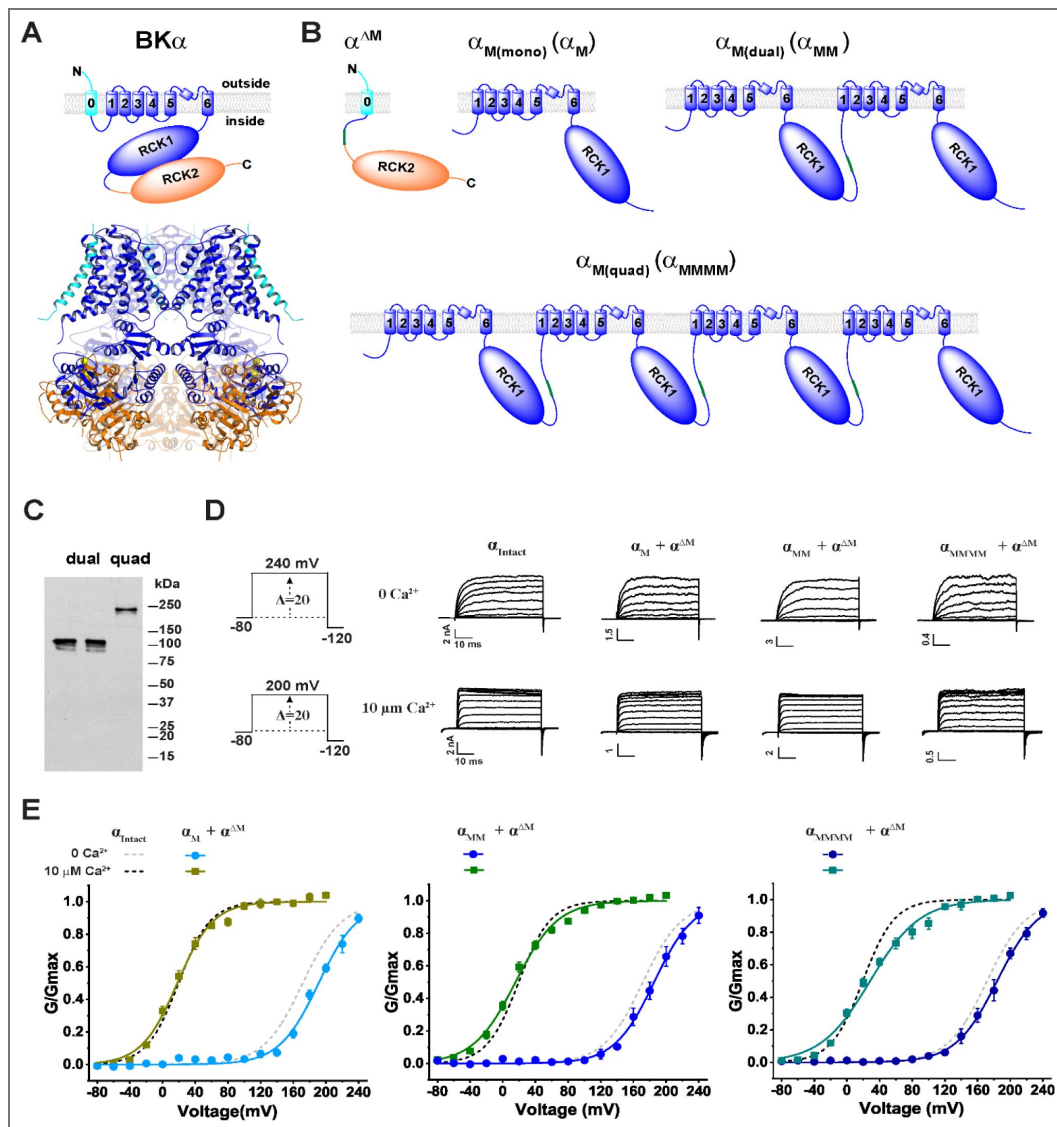


Figure 1. Design of concatenated BK α subunit constructs that form functional channels.

(A) Schematic of the membrane topology and side-view of the tetrameric 3D structure of the BK α subunit (PDB ID: 8GHF; cryo-EM structure in plasma membrane (3)) highlighting the three complementary separable regions in different colors. For clarity, the front and back subunits are shown in a partially transparent mode. (B) Schematic of the membrane topology for the BK α_M module, concatenated dual- and quadruple-repeat constructs, and the complementary BK $\alpha^{\Delta M}$ construct. (C) Immunoblot analysis of the V5-tagged $\alpha_{M(\text{dual})}$ (left, α_{M1M2} ; right, α_{M3M4}) and BK $\alpha_{M(\text{quad})}$ constructs transiently expressed in HEK293 cells with an anti-V5 antibody. (D) Representative current traces from BK channels formed by intact, single repeat BK α_M , dual-repeat BK $\alpha_{M(\text{dual})}$, and quadruple-repeat BK $\alpha_{M(\text{quad})}$ constructs in response to membrane depolarization from -80 mV in 20-mV steps at 0 and 10 μM intracellular free Ca^{2+} . (E) Voltage dependence of BK channel activation for channels formed by the single (left), dual (middle), and quadruple (right) α_M constructs in the absence and presence of 10 μM Ca^{2+} . Electrophysiological recordings were repeated $n = 4-10$, as indicated in Table 1. Error bars represent $\pm$ SEM.

Table 1. Boltzmann-fit parameters of the voltage-dependent concatenated tandem BK channel activation in the wildtype, mutants in the absence and presence of intracellular Ca^{2+} .

Expression*	Ca^{2+} (μM)	Boltzmann fit parameters		
		$V_{1/2}$ (mV)	z	n [§]
α_{intact}	0	172 ± 2	1.08 ± 0.04	9
α_{intact}	10	19 ± 4	1.47 ± 0.10	8
α_{M}	0	191 ± 1	1.11 ± 0.13	4
α_{M}	10	18 ± 3	1.20 ± 0.06	6
α_{MM}	0	186 ± 5	1.12 ± 0.12	5
α_{MM}	10	16 ± 3	1.02 ± 0.03	10
α_{MMMM}	0	184 ± 5	1.09 ± 0.07	6
α_{MMMM}	10	29 ± 4	0.86 ± 0.06	4
$\alpha_{\text{intact}} + \gamma 1$	0	35 ± 2	1.56 ± 0.11	8
$\gamma 1 \alpha_{\text{MM}}$	0	56 ± 4	0.99 ± 0.07	6
$\gamma 1 \alpha_{\text{MMMM}}$	0	42 ± 3	1.13 ± 0.13	5
$\alpha_{\text{M}}^{\text{L312A}}$	0	62 ± 4	0.96 ± 0.07	6
$\alpha_{\text{M}}^{\text{L312A}}_{\text{M}}^{\text{WT}}$	0	121 ± 6	0.90 ± 0.04	10
$\alpha_{\text{M}}^{\text{WT}}_{\text{M}}^{\text{L312A}}$	0	118 ± 5	0.95 ± 0.07	4
$\alpha_{\text{M}}^{\text{L312A}}_{\text{M}}^{\text{L312A}}$	0	64 ± 5	0.97 ± 0.08	5
$\alpha_{\text{M}}^{\text{L312A}}_{\text{M2}}^{\text{WT}}_{\text{M3}}^{\text{WT}}_{\text{M4}}^{\text{WT}}$	0	153 ± 4	0.96 ± 0.10	5
$\alpha_{\text{M}}^{\text{L312A}}_{\text{M}}^{\text{L312A}}_{\text{M}}^{\text{WT}}_{\text{M}}^{\text{WT}}$	0	133 ± 2	0.77 ± 0.05	4
$\alpha_{\text{M}}^{\text{L312A}}_{\text{M}}^{\text{L312A}}_{\text{M}}^{\text{L312A}}_{\text{M}}^{\text{WT}}$	0	98 ± 5	0.90 ± 0.11	5
$\alpha_{\text{M}}^{\text{L312A}}_{\text{M}}^{\text{L312A}}_{\text{M}}^{\text{L312A}}_{\text{M}}^{\text{L312A}}$	0	61 ± 6	1.05 ± 0.09	5
$\alpha_{\text{M}}^{\text{L312A}}_{\text{M}}^{\text{L312A}}_{\text{M}}^{\text{L312A}}_{\text{M}}^{\text{L312A}} + \alpha_{\text{M}}^{\text{WT}}_{\text{M}}^{\text{WT}}_{\text{M}}^{\text{WT}}_{\text{M}}^{\text{WT}}$	10	180 ± 13 (36%)	0.90 ± 0.52	5
		65 ± 8 (64%)	1.01 ± 0.13	
$\alpha_{\text{intact}}^{\text{V288A}}$	10	149 ± 8	1.06 ± 0.10	5
$\alpha_{\text{M}}^{\text{V288A}}_{\text{M}}^{\text{WT}}$	10	31 ± 2	1.14 ± 0.09	5
$\alpha_{\text{M}}^{\text{V288A}}_{\text{M}}^{\text{WT}}_{\text{M}}^{\text{WT}}_{\text{M}}^{\text{WT}}$	10	21 ± 11	1.16 ± 0.06	3
$\alpha_{\text{M}}^{\text{V288A}}_{\text{M}}^{\text{V288A}}_{\text{M}}^{\text{WT}}_{\text{M}}^{\text{WT}}$	10	19 ± 2	1.19 ± 0.17	3
$\alpha_{\text{M}}^{\text{V288A}}_{\text{M}}^{\text{V288A}}_{\text{M}}^{\text{V288A}}_{\text{M}}^{\text{WT}}$	10	38 ± 3	1.05 ± 0.06	3
$\alpha_{\text{M}}^{\text{V288A}}_{\text{M}}^{\text{V288A}}_{\text{M}}^{\text{V288A}}_{\text{M}}^{\text{V288A}}$	10	127 ± 10	0.75 ± 0.21	3

* Except the full-length BK α_{intact} construct, all other listed (derivatives of α_{M}) constructs were co-expressed with the complementary $\alpha^{\Delta\text{M}}$ -GFP construct.

§ The number of recorded excised inside-out patches from different HEK293 cells.

A single LRRC26 ($\gamma 1$) subunit per channel is sufficient to fully modulate BK channels

The $\gamma 1$ subunit's “all-or-none” modulatory effect on BK channels has remained mechanistically elusive, despite the recent availability of 3D structures of the BK $\alpha/\gamma 1$ complexes. To directly investigate the functional stoichiometry of BK channel modulation by the $\gamma 1$ subunit, we fused the C-terminus of the $\gamma 1$ subunit to the N-terminus of the first α_M repeat in the α_{MM} and α_{MMMM} constructs, generating the BK $\gamma 1\alpha_{MM}$ (BK $\gamma 1\alpha_{M(dual)}$) and BK $\gamma 1\alpha_{MMMM}$ (BK $\gamma 1\alpha_{M(quad)}$) fusion constructs (Fig. 2B). Immunoblot analysis of the V5-tagged constructs using an anti-V5 antibody showed a major band at the expected size for both BK $\gamma 1\alpha_{MM}$ (~150 kDa) and BK $\gamma 1\alpha_{MMMM}$ (~270 kDa) (Fig. 2C), confirming proper expression and stability. Upon co-expression of these constructs with α^{AM} -GFP in HEK293 cells, we observed that the N-terminally fused $\gamma 1$ subunit induced a $V_{1/2}$ shift of 125 mV with the BK $\gamma 1\alpha_{MM}$ construct ($V_{1/2} = 62 \pm 5$ mV) and 145 mV with the BK $\gamma 1\alpha_{MMMM}$ construct ($V_{1/2} = 38 \pm 4$ mV) in the virtual absence of Ca^{2+} (Fig. 2D, E). These large $V_{1/2}$ -shifting effects in stoichiometrically defined $\gamma 1:\alpha = 1:2$ and $1:4$ channel complexes clearly indicate that a single $\gamma 1$ subunit per tetrameric channel is sufficient to fully modulate BK channels. This “one-subunit-sufficient” effect provides a mechanistic explanation for the observed binary, all-or-none modulation by the $\gamma 1$ subunit when its expression is limited.

L312A mutation reveals stoichiometrically graded gating and confirms functional integrity of concatenated constructs

To ensure that the concatenated α_M constructs are suitable for studying the stoichiometry of BK channel gating and regulation, the constructs must demonstrate two additional aspects of functional integrity beyond simply forming functional channels. First, each α_M repeat within the construct should contribute equally to channel formation. Second, the concatenated dual and quadruple α_M repeats in the $\alpha_{M(dual)}$ and $\alpha_{M(quad)}$ constructs should assemble primarily as complete units, i.e., two $\alpha_{M(dual)}$ constructs or one $\alpha_{M(quad)}$ construct should form a single channel. The latter ensures that heterogeneous channels with uncontrolled subunit stoichiometry are unlikely to form and confound the results.

To evaluate the functional integrity of the concatenated constructs and to examine the stoichiometry of activation gating in the PGD, we investigated the mutational effects of the deep pore residue L312 (³⁰⁷FILGGLAMFAS³¹⁷) (Fig. 3A), which plays a pivotal role in BK channel activation gating. Most mutations at this site resulted in constitutively active channels (41). The L312A mutation, in particular, causes a substantial shift of the $V_{1/2}$ toward hyperpolarization potentials in the absence of Ca^{2+} (41, 42). We first confirmed that the L312A mutation on the single-repeat α_M construct caused a similarly large (~130 mV) shift in $V_{1/2}$ toward hyperpolarization in Ca^{2+} -free conditions (Fig. 3B, C; Table 1). We then introduced the L312A mutation into the dual- and quadruple-repeat α_{MM} and α_{MMMM} constructs in a repeat-specific manner. For the α_{MM} construct, introducing L312A mutation into either the first or second repeat resulted in a similar shift (65 or 68 mV) in $V_{1/2}$, approximately half of the total shift seen when both repeats (i.e., all subunits) were mutated (Fig. 3C; Table 1). For the α_{MMMM} construct, we observed that the voltage dependence of the channel activation shifted progressively with each additional L312A mutation introduced. Specifically, the initial L312A mutation on the first repeat shifted $V_{1/2}$ by 31 mV, with further increases of 20, 35, and 37 mV in $V_{1/2}$ shifts observed for the additional mutation on the second, third, and fourth repeats, respectively (Fig. 3D; Table 1). The L312A mutation also caused a slowed decay in tail currents at negative voltages (Fig. 3B) compared to unmutated channels (Fig. 1D). Fitting the current decay kinetics revealed that the time constant (τ) of the decay was increased with the number of L312A mutations present in the channels formed by α_{MM} and α_{MMMM} constructs (Fig. 1E). It is worth noting that the tail current decay rates for channels with zero or one L312A mutation were likely overestimated due to rapid closure at very negative voltages (-120 mV) in Ca^{2+} -free conditions, exceeding the detection limit of the 2 kHz-filtered recordings. These results demonstrate a stoichiometrically incremental effect of the

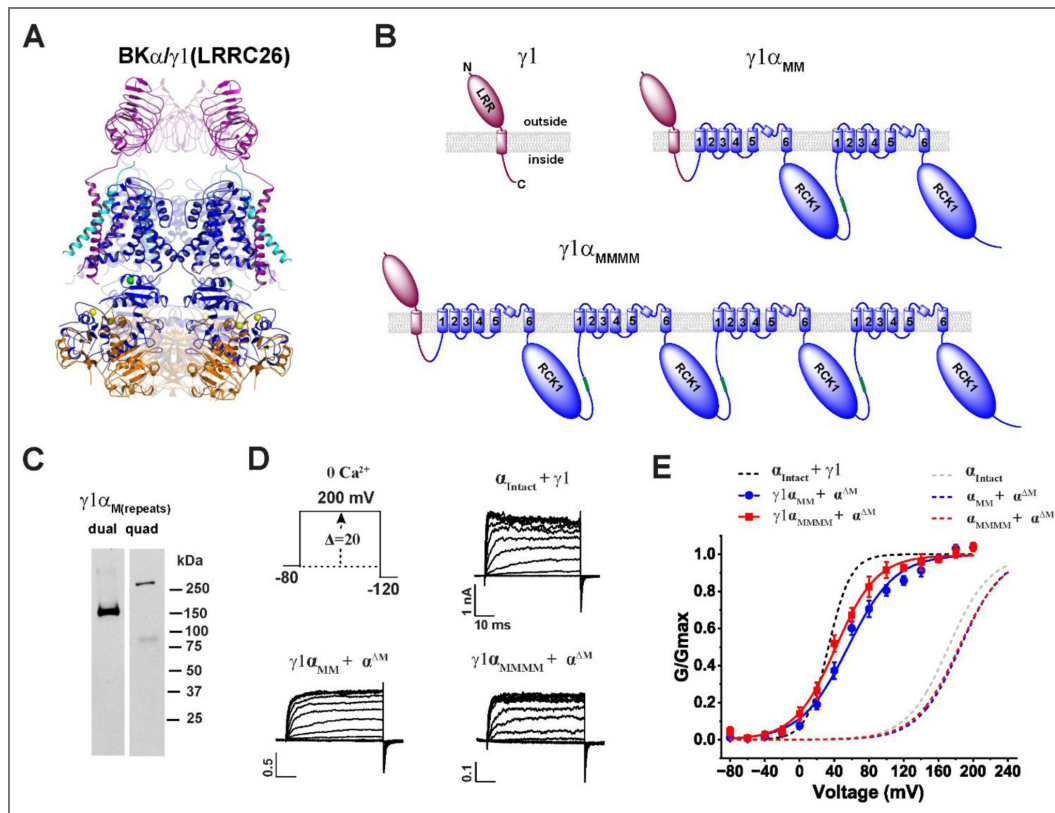


Figure 2. A single $\gamma 1$ subunit per BK channel is sufficient for full modulation.

(A) Side-view of the 3D structure of the BK α / $\gamma 1$ channel complex (PDB ID: 7Y03 (23)) showing the $\gamma 1$ subunit in purple and the three separable BK α regions in distinct colors. **(B)** Schematic of the membrane topology for the $\gamma 1$ subunit and its fusion constructs, created by linking its C-terminus to the N-terminus of the α_M dual- and quadruple-repeat constructs. **(C)** Immunoblot analysis of the V5-tagged $\gamma 1\alpha_{M(\text{dual})}$ and $\gamma 1\alpha_{M(\text{quad})}$ constructs expressed in HEK293 cells with an anti-V5 antibody. **(D)** Representative current traces from BK channels formed by co-expressing the $\gamma 1$ subunit with the intact BK α or by $\gamma 1$ -fusion to the concatenated α_M constructs (co-expressed with $\alpha^{\Delta M}$) in response to membrane depolarization from -80 mV in 20-mV steps in the virtual absence of Ca^{2+} . **(E)** Voltage dependence of activation for channels formed by the $\gamma 1\alpha_{M(\text{dual})}$ and $\gamma 1\alpha_{M(\text{quad})}$ constructs co-expressed with $\alpha^{\Delta M}$. Electrophysiological recordings were repeated $n = 5 - 8$, as indicated in Table 1. Error bars represent $\pm$ SEM.

L312A mutation on BK channel voltage gating. Importantly, they also indicate that individual $BK\alpha_M$ repeats in both the dual- and quadruple-repeat constructs contribute similarly to the gating properties of the assembled channels.

The conductance-voltage (G-V) curves of BK channels formed by these L312A mutant constructs were well-fit by a single Boltzmann function, with slopes (i.e., apparent gating charge z) remaining within the typical range (Fig. 3C, D; Table 1), consistent with a largely homogenous channel population. In contrast, G-V curves of the channels formed by co-transfecting cells with WT and L312A single $BK\alpha$ subunit constructs (1:1 DNA ratio) showed shallow slopes (Fig. 3F), agreeing with the predicted heterogeneity in subunit composition and $V_{1/2}$ values. To confirm that each $\alpha_{M(\text{quad})}$ construct predominantly forms a single channel without subunit exchange, we co-expressed fully mutated α_M^{L312A} and unmutated α_M^{WT} constructs (1:1 DNA ratio). The resulting G-V curves were best fit with a double-Boltzmann function showing $V_{1/2}$ values matching those of the all-WT and all-L312A channels (Fig. 3F; Table 1). Additionally, tail currents at -120 mV exhibited two distinct exponential decay components differing by ~10-fold in rate, corresponding to populations of all-WT and all-mutated channels (Fig. 2G). These findings support that the concatenated α_M repeats in the $\alpha_{M(\text{dual})}$ and $\alpha_{M(\text{quad})}$ constructs function as a whole unit in channel assembly, enabling stoichiometrically defined investigation of channel gating and regulation.

V288A triggers selectivity filter inactivation through an all-subunit mechanism

The selectivity filter in K^+ channels is directly involved in C-type inactivation. Given its potential role in BK channel activation gating (43, 44), we examined the subunit stoichiometric effects of structural perturbations within the selectivity filter on BK channel gating. We found that the V288A mutation, located within the K^+ -selective signature sequence ($^{286}\text{STVGYGD}^{292}$) (Fig. 3A), produced profound effects on BK channel gating. We previously reported that mutations near the selectivity filter, e.g., in the P-helix (Y279) or at the extracellular side of the filter (Y294), can induce an atypical closed-state-coupled C-type inactivation in BK channels under low extracellular K^+ conditions (44). Interestingly, V288A, even without reduced extracellular K^+ , induced a similar slow inactivation process, causing a gradual decrease in the availability of activatable channels under conditions (e.g., negative voltages) that promote channel closure (Fig. 4A). Consequently, compared to WT channels, V288A mutant channel currents developed very slowly in response to depolarization (Fig. 4B). However, following a long pre-depolarization, the mutant channels fully recovered from the inactivated state and behaved similarly to WT channels in the kinetics and voltage-dependence of activation (Fig. 4C).

Using the concatenated $BK\alpha_M$ constructs, we investigated the subunit stoichiometry of V288A-induced gating effects by introducing the mutation into different numbers of $BK\alpha$ subunits within a single channel. With the $BK\alpha_{M(\text{dual})}$ construct, we generated α_M^{V288A} and α_M^{V288A} constructs, harboring the mutation on half or all subunits, respectively. We introduced one (α_M^{V288A}), two (α_M^{V288A}), three (α_M^{V288A}), four (α_M^{V288A}) mutations on the $BK\alpha_{M(\text{quad})}$ construct. Interestingly, unlike the one-subunit-sufficient effect of LRRC26 or the stoichiometrically incremental effects of the L312A mutation, V288A exerted an “all-subunit-required” modulatory effect: the V288A-induced changes occurred only when all four $BK\alpha$ subunits in a channel were mutated. Channels partially mutated, originated from α_M^{V288A} , α_M^{V288A} , α_M^{V288A} , or α_M^{V288A} , showed no significant differences from WT channels in the time course (Fig. 4D, E) or voltage-dependence (Fig. 4F) of depolarization-induced currents. In contrast, channels with all α_M repeats mutated (α_M^{V288A} and α_M^{V288A}) displayed markedly slowed current development, approximately 100-fold slower than WT (Fig. 4D, E). These fully mutated channels also showed an apparently higher $V_{1/2}$ in their G-V relationships (Fig. 4F). However, since the V288A mutation has no significant effect on the channel’s normal activation gating (Fig. 4C, F), this apparent shift in G-V curves likely resulted from incomplete recovery of inactivated channels during 50 ms depolarization phase in the

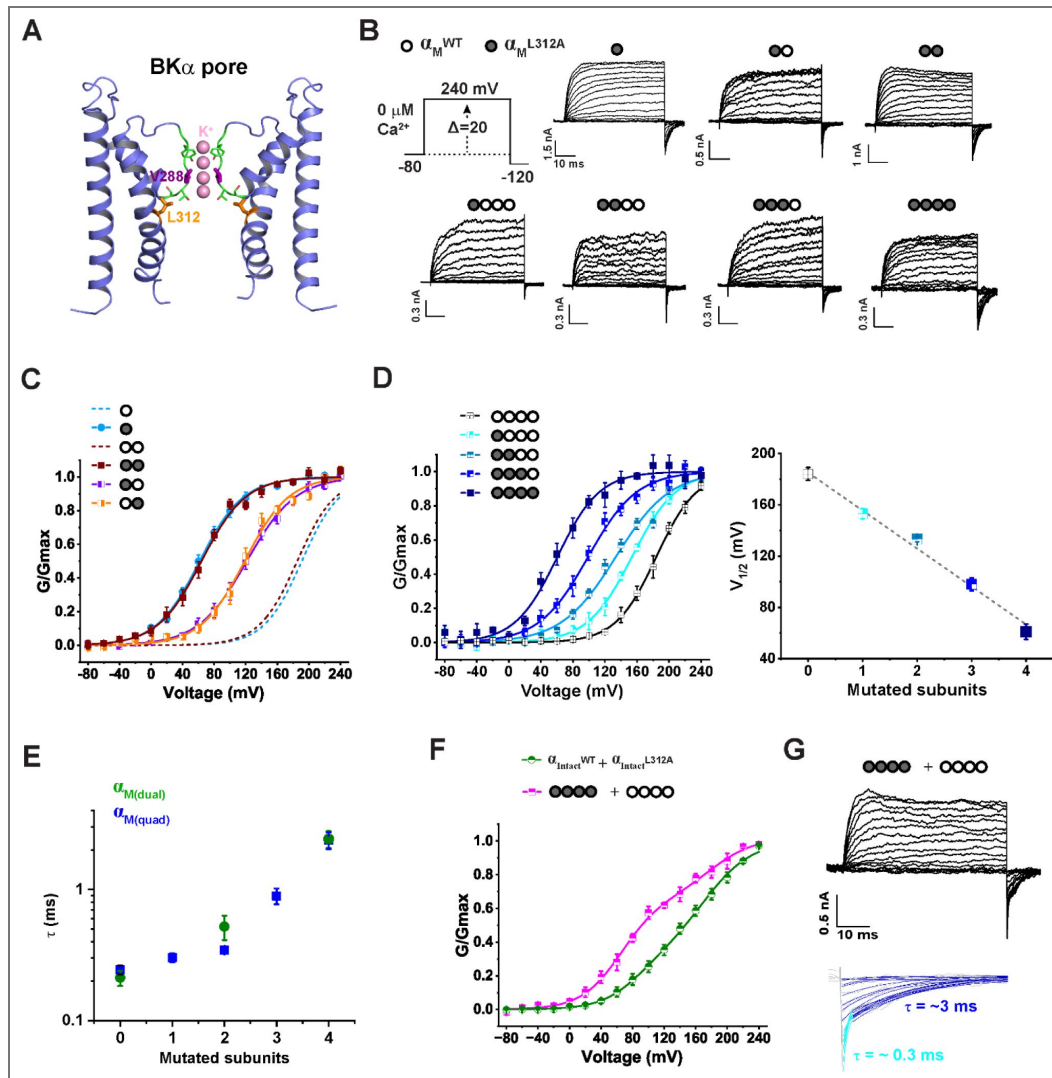


Figure 3. Stoichiometrically incremental effect of the L312A mutation on BK channel voltage gating and validation of functional integrity of the concatenated constructs.

(A) Side-view of the BK channel pore structure (PDB ID: 8GHF (3)), highlighting the deep pore residue L312 and selectivity filter residues (stick and line modes). Only two diagonal pore-domains are shown for clarity. (B) Representative current traces from BK channels formed by single α_M and concatenated $\alpha_{M(dual)}$ and $\alpha_{M(quad)}$ constructs containing subunit-specific L312A mutations. Depolarizations from -80 mV were applied in 20 -mV steps. Mutated subunits are indicated by filled circles and WT subunits by empty circles. (C) Voltage dependence of activation for channels formed by the indicated L312A mutant α_M and $\alpha_{M(dual)}$ constructs co-expressed with α^{AM} . Dashed lines show G-V curves of the corresponding non-mutated channels for comparison. (D) Voltage dependence of activation for channels formed by the $\alpha_{M(quad)}$ constructs with different numbers of L312A mutations. A plot of the $V_{1/2}$ vs. the number of mutated subunits is shown. (E) Plot of tail current decay rates (-120 mV) vs. number of L312A mutations, from $\alpha_{M(dual)}$ and $\alpha_{M(quad)}$ constructs. (F) Voltage dependence of activation for channels formed by co-expression of WT and L312A-mutant intact BK α subunits ($n = 6$), or non-mutated and fully mutated $\alpha_{M(quad)}$ constructs co-expressed with α^{AM} . (G) Representative current traces from channels formed by co-expressing the non-mutated and fully L312A mutated $\alpha_{M(quad)}$ constructs (co-expressed with α^{AM}). Enlarged and fitted tail currents are shown below. Electrophysiological recordings were performed under Ca^{2+} -free conditions and repeated $n = 4 - 10$ as indicated in Table 1. Error bars represent $\pm$ SEM.

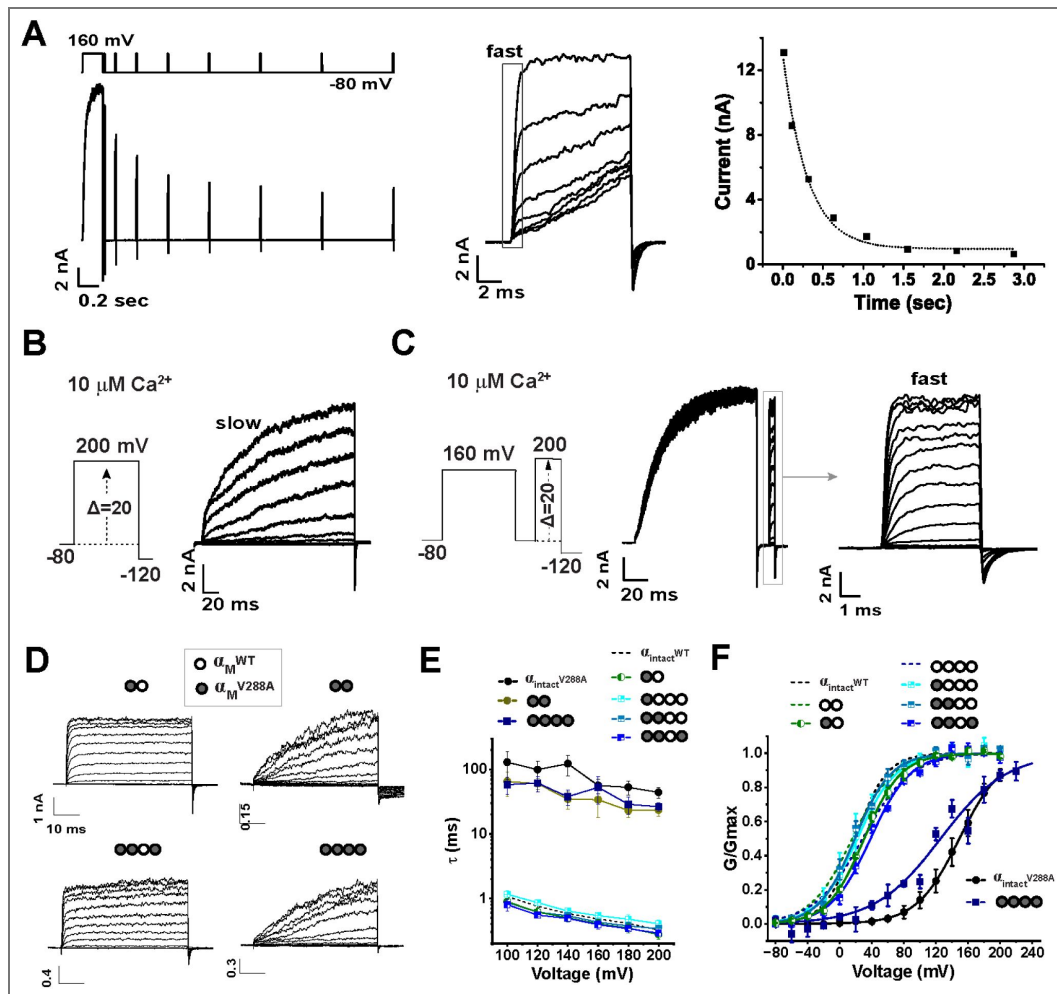


Figure 4. V288A-induced selectivity filter inactivation requires mutation of all subunits.

(A) Time-dependent inactivation of V288A-mutant BK α (intact) channels at -80 mV after a prolonged depolarization (160 mV for 200 ms). Inactivation was assayed by monitoring the reduction in fast-activating currents (shown and compared in the middle) elicited by brief depolarization (160 mV for 10 ms) after a prior -80 mV holding time of 10 ms, 0.1 s, 0.2 s, 0.3 s, 0.4 s, 0.5 s, 0.6 s, and 0.7 s. The amplitudes of fast-activating currents are compared (middle) and plotted against time (right). (B) Representative current traces of V288A mutant BK α (intact) channels showing slowly developing depolarization-induced currents. (C) V288A mutant channel exhibited normal activation gating following recovery (160 mV for 100 ms) from inactivation, as indicated by currents elicited by brief depolarization to different voltages after brief repolarization. (D) Representative current traces from channels formed by concatenated $\alpha_{M(dual)}$ and $\alpha_{M(quad)}$ constructs with subunit-specific V288A mutations (co-expressed with $\alpha_{M^{WT}}$). Mutant and WT subunits are indicated as filled and empty circles, respectively. (E) Depolarization-induced current development rates for channels formed by non-mutated and V288A-mutant BK α (intact) and concatenated $\alpha_{M(dual)}$ and $\alpha_{M(quad)}$ constructs. Electrophysiological repeats: $n = 8$ for $\alpha_{(intact)}^{WT}$, 5 for $\alpha_{(intact)}^{V288A}$, 4 for $\alpha_{M^{V288A}}^{WT}$, 4 for $\alpha_{M^{V288A}}^{V288A}$, 4 for $\alpha_{M^{V288A}}^{WT}$, 4 for $\alpha_{M^{V288A}}^{V288A}$, 4 for $\alpha_{M^{V288A}}^{WT}$, 4 for $\alpha_{M^{V288A}}^{V288A}$, 3 for $\alpha_{M^{V288A}}^{WT}$, 3 for $\alpha_{M^{V288A}}^{V288A}$, and 4 for $\alpha_{M^{V288A}}^{WT}$, 4 for $\alpha_{M^{V288A}}^{V288A}$. (F) Voltage dependence of depolarization-induced currents for BK channels formed by non-mutated and V288A-mutant BK α (intact) and concatenated $\alpha_{M(dual)}$ and $\alpha_{M(quad)}$ constructs. Electrophysiological repeats $n = 3-5$ as indicated in Table 1. All recordings were performed using symmetric K $^{+}$ (140 mM) solutions with 10 μ M intracellular Ca $^{2+}$. Error bars represent $\pm$ SEM.

voltage protocol used. Together, these results demonstrate that V288A induces an “all-subunit-required” inactivation process at the selectivity filter, while the activation/deactivation processes of normal channel gating remain largely unchanged.

Discussion

Concatenated constructs have been extensively used to study ion channel subunit stoichiometry (32–38), enabling a deeper understanding of the mechanisms underlying channel gating and regulation by homo- or heteromeric subunits, voltage, ligands, and mutations. However, the BK channel is one of the few channels whose principal subunits have their N-termini located on the extracellular side. This unique membrane topology prevents the direct application of the traditional N-to-C-terminal concatenation method for generating the concatenated constructs. In this study, we employed a strategy that involved splitting and fusing BK α subunits into two modular constructs that reconstitute functional BK channels. We validated the functionality of these concatenated constructs by demonstrating that the resulting channels closely resemble intact BK channels in their voltage and Ca²⁺ dependence of activation. Furthermore, we confirmed that each repeat with the constructs contributes similarly to channel function, as evidenced by the stoichiometrically incremental effect of the L312A mutation on voltage gating. Previous studies have reported that, in some cases, subunits from different quadruple-repeat concatamers can assemble aberrantly, leading to the formation of mixed channels with altered properties (45, 46). The large functional effect of the L312A mutation on BK channel gating allowed us to distinguish between properly assembled channels and potential inter-subunit crossover products. Our results showed that the concatenated tandem constructs assemble predominantly as intended, i.e., each channel is formed by two dual-repeat constructs or a single quadruple-repeat construct, as evidenced by the absence of significant heterogeneity in channel gating properties. Using these well-defined concatenated constructs, we identified three distinct types of subunit stoichiometry in BK channel gating or modulation. These rule out the possibility that the observed stoichiometric effects are artifacts arising from the construct design or assembly defects. Thus, we have demonstrated that our engineered concatenated BK α constructs serve as effective molecular tools for probing the subunit stoichiometry in BK channel gating and regulation.

Most auxiliary proteins of the K⁺ channels exhibit stoichiometrically incremental effects on channel modulation. However, the auxiliary γ 1 subunit displays an unusual binary “all-or-none” modulatory effect on BK channels (19, 25), despite being able to bind to BK α subunits at a 1:1 molecular ratio (29–31). In recently reported cryo-EM structures of channel complexes, the four γ 1 subunits display an apparent four-fold symmetry in TM domain interactions with their LRR domains tetramerized on the extracellular side (22–24). These structural features raise the possibility that the four γ 1 subunits might act collectively in BK channel modulation (22), which appears to contradict the “one-subunit-sufficient” mechanism previously inferred from single channel gating properties of BK channels modulated by a β 2- γ 1 chimeric construct (29). With the concatenated modular BK α constructs developed in this study, we were able to directly control the subunit stoichiometry of γ 1 subunits relative to BK α subunits. The intracellular location of the N-terminus in the concatenated BK α constructs enabled us to fuse the γ 1 subunit to the N-terminal side of the α_M module. With γ 1-fused concatenated α_M dual- and quadruple-repeat constructs, we provide direct evidence that one and two copies of the γ 1 subunits per BK α tetramer are sufficient to produce the full modulatory effect of the γ 1 subunit on BK channel gating. Thus, our findings unequivocally confirm the “one-subunit-sufficient” mechanism of the γ 1 subunit in BK channel modulation by using the full length γ 1 subunit, stoichiometrically defined BK α / γ 1 channel complexes, and macroscopic currents from large channel populations. This result, combined with the structural observation of symmetrical binding of γ 1 to all four voltage-sensor domains, raises the intriguing possibility that γ 1 modulation may occur through asymmetric allosteric coupling, despite symmetric structural binding—a phenomenon warranting further mechanistic investigation.

Through conformational linkage to the voltage- and Ca^{2+} -sensors, the movement of the lower half of the pore-lining S6 helix ultimately controls BK channel pore gating. According to the Horrigan-Aldrich gating model (4), BK channel activation involves a rate-limiting process of pore-gate opening, regulated by four independent and identical voltage- and Ca^{2+} -sensors. However, the subunit stoichiometry underlying the S6 movement-induced pore-gate opening in BK channels remains unclear and warrants investigation. The deep pore residue L312 on S6 is unique in that it lies adjacent to the double-glycine gating hinge residues (G310 and G311), is positioned immediately below the selectivity filter (Fig. 3A), and appears to be the most mutation-sensitive residue affecting channel activation gating, as most of its substitution mutations result in constitutively open channels (41). Thus, L312 appears to represent a structural endpoint for the voltage- and Ca^{2+} -induced conformational changes in S6 and plays an essential role in stabilizing the channel's closed state. In classic Shaker K^+ channels, a mutation of the neighboring glycine hinge residue (G466 in Shaker, corresponding to G311 in BK α) in concatenated tetrameric constructs was reported to display a concerted, “one-subunit-sufficient” effect, where all mutant subunit combinations produced similar effects on channel gating (37), a phenomenon reminiscent of the modulatory behavior of the LRRC26 (γ 1) subunit on BK channel gating. In contrast, our current study reveals a stoichiometrically graded (independent) effect of the L312A mutation on BK channel voltage gating. Similarly, single channel recordings showed that a mutation to arginine at the neighboring G311 residue also produced an additive effect on the BK channel voltage-gating proportional to the number of mutated subunits (47). Together, the observed stoichiometric independence of mutational effects at the L312 and G311 residues is consistent with the modeled independence of individual voltage and Ca^{2+} sensors in channel activation (4, 47), and likely reflects a fundamental difference in the location and mechanism of the activation gate. Currently, the fundamental question of pore gate location in BK channels remains unsettled. In most K^+ channels, a hydrophobic “bundle-crossing” gate near the intracellular end of the pore controls channel activation (48). However, BK channels appear to lack such a gate, as the pore remains structurally wide open in the presumed closed-state (Ca^{2+} -free) structures (2, 8) and is readily accessible to large intracellular blockers (49, 50) or cysteine-modifying reagents (51) even when the channel is closed. Thus, in Shaker and related Kv channels, the classical activation gate resides within the lower S6 helices, where concerted subunit movements are required for highly cooperative transitions from closed to open states. In contrast, the graded, additive effects of deep pore mutations in BK channels provide evidence that no such concerted gating structure exists within the lower S6 or at least not within the deep pore region.

The selectivity filter, known to govern C-type inactivation in many channels (52–54), may also serve as activation gate in some, such as the ligand-gated CNG channels (55). Accordingly, direct involvement of the selectivity filter in BK channel activation gating has long been speculated (34, 43, 49), but compelling experimental validation remains lacking. While C-type inactivation doesn't normally occur in BK channels, we previously found that it can be induced by mutations near the selectivity filter in combination with low extracellular K^+ (44). Interestingly, the induced C-type inactivation in BK channels is closed-state coupled (44), opposite to the open-state coupled C-type inactivation commonly observed in other channels. In this study, we report that the V288A mutation, located within the K^+ -selective signature sequence, also induces pronounced inactivation even under normal extracellular K^+ conditions. Our analysis of the subunit stoichiometric effects of V288A using concatenated BK α dual- and quadruple-repeat constructs clearly showed that modifications in all four subunits are required to elicit the mutation-induced inactivation, whereas all other mutant subunit combinations produced minimal effects on BK channel gating. It is noteworthy that the stoichiometric effects of mutations on the rate of C-inactivation varies depending on the location of the mutated residue in other voltage-gated K^+ channels. Mutations located more peripherally from the selectivity filter signature motif often produced either a non-linear, graded effect consistent with a cooperative inter-subunit interaction (e.g., A463V, A449K, and W434F in Shaker channels, or S631A and T618A in hERG1) (36, 38, 56), or a linearly additive effect consistent with inter-subunit independence (e.g., M645C in hERG1) (38). In contrast, mutations near or within the selectivity filter signature motif affected C-type inactivation in an all-or-none manner, with a single subunit mutation, such as S620T or G628C in hERG1, being

sufficient to eliminate or attenuate inactivation to the same extent as observed in mutant homotetramers (38). This effect closely mirrors our observations with V288A in BK channels, where mutations in all four subunits are required to induce and maintain inactivation, supporting our interpretation that V288A induces a form of C-type inactivation, similar to what we previously observed with mutations at the extracellular mouth and the P-helix (44). Given its critical location, the V288A mutation may alter BK channel ion selectivity, as an alanine substitution at the equivalent position in Kv4.3 has been shown to reduce K⁺ selectivity (57). Whether V288A induces both C-type inactivation and changes in ion selectivity through a similar subunit stoichiometry remains to be determined.

The contrasting behaviors of L312A and V288A mutations in concatenated BK α_M constructs reveal distinct subunit stoichiometry requirements between the deep pore activation gating and selectivity filter inactivation gating. A typical K⁺ channel activation gate requires all four subunits undergo a concerted conformational change to ensure an all-or-none transition between a fully conductive and non-conductive pore (58, 59), a hallmark of classic voltage-gated K⁺ channels that allows rapid and precise control of membrane excitability. Such tight inter-subunit all-or-none cooperativity observed at the BK channel selectivity filter makes it a plausible candidate for serving as the activation gate, a property not yet demonstrated for the lower S6 segment. It is possible that the selectivity filter functions as the physical gate for two gating processes, activation and inactivation, that can occur on different timescales and through distinct structural mechanisms. Previously, we found that the voltage- and Ca²⁺-dependence of inactivation and activating gating were well correlated in BK channels (44), suggesting a shared or coupled energetic pathway. L312 lies in close proximity to and physically interacts with the selectivity filter, providing a possible structural link for transmitting energy from S6 movement to the selectivity filter during activation gating. Further investigation of inactivation and its relationship to activation will likely help elucidate the role of the selectivity filter in BK channel activation gating.

In conclusion, our study employed an innovative strategy to generate concatenated subunit constructs and investigate the subunit stoichiometry and modulation of BK channels. The development of these constructs enabled detailed exploration of the intricate gating and regulatory mechanisms of BK channels in a stoichiometrically subunit-specific manner. Using these concatenated constructs, we identified three distinct types of subunit stoichiometry in BK channel modulation: an additive (independent) type and two contrasting all-or-none types, namely, “one-subunit-sufficient” and “all-subunit-required”. These represent divergent stoichiometric modes of gating control by the pore, LRRC26 (γ 1), and selectivity filter, respectively. This study offers new molecular tools and advances our understanding of subunit stoichiometry in BK channel gating and modulation.

Materials and Methods

Generation of concatenated tandem BK α repeat constructs and expression of BK channels

We first generated a pcDNA6-based plasmid, pcDNA6-myc-BK α -V5-His, carrying *KCNMA1* cDNA (GenBank: U11058). This plasmid expresses the full-length (1113 amino acids) human BK α (GenBank: AAB65837) with an N-terminal Myc tag and C-terminal V5 and 6 $\times$ His tags, serving as the template for further plasmid constructions. To express BK α 's main region (residues 44–651) as a protein module, we created pcDNA6-myc- α_M -V5-His by deleting the nucleotide sequences encoding N-terminal residues 1–43 (extracellular N-terminus and S0 TM segment) and C-terminal residues 652–1113 (RCK2 domain and C-terminal tail). Next, we constructed a complementary plasmid, pcDNA6-myc-BK $\alpha^{\Delta M}$ -GFP-V5-His, by replacing residues 94–651 of BK α with a flexible peptide linker (SSGGGSGGGSGGAR) and tagging monomeric enhanced GFP to the C-terminus. This complementary plasmid enables functional channel formation when co-expressed with a plasmid encoding a single α_M or concatenated α_M repeats. To construct structurally stable plasmids expressing concatenated α_M repeats, we synthesized three codon-optimized DNA sequences

encoding the same α_M module, each differing by $\sim 25\%$ in nucleotide sequence from each other and the original *KCNMA1* cDNA. Using these synthesized sequences, we constructed the dual-repeat expressing plasmid, pcDNA6- $\alpha_{M1}\alpha_{M2}$ -V5-His, in which α_{M1} (residues 43-649) is preceded by a short initiation sequence (MGS) and linked to α_{M2} (also residues 43-649) via a flexible linker (GGGGSGSAG). A NotI restriction site with a peptide spacer (GGGKPIPNAAG) was inserted between α_{M2} and the V5 tag. We also generated a second dual-repeat expressing plasmid, the pcDNA6- $\alpha_{M3}\alpha_{M4}$ -V5-His, in which α_{M3} (residues 44-649) is preceded by an N-terminal sequence (MGAAAA) containing a NotI site and linked α_{M4} (residues 43-649) via a flexible linker (GGGSAAGSG). As the two dual-repeat constructs produce highly similar proteins and exhibit no difference in electrophysiological properties, both are referred to as BK $\alpha_{(dual)}$ or α_{MM} . To generate a quadruple-repeat expressing plasmid, pcDNA6- $\alpha_{M1}\alpha_{M2}\alpha_{M3}\alpha_{M4}$ -V5-His, we subcloned the $\alpha_{M3}\alpha_{M4}$ dual-module fragment from pcDNA6- $\alpha_{M3}\alpha_{M4}$ -V5-His into pcDNA6- $\alpha_{M1}\alpha_{M2}$ -V5-His using NotI and AgeI (located between the V5 and 6 $\times$ His tags). The expressed protein is referred as BK $\alpha_{(quad)}$ or α_{MMMM} . For $\gamma 1$ (LRRC26) fusion constructs, we generated pcDNA6-BK $\gamma 1\alpha_{M1}\alpha_{M2}$ -V5-His and by inserting the $\gamma 1$ (LRRC26) sequence at the N-terminus of $\alpha_{M1}\alpha_{M2}$ (pcDNA6- $\alpha_{M1}\alpha_{M2}$ -V5-His) via a 16-residue flexible linker (SSGSGSESKSTGGSGS). The expressed fusion protein is designated as BK $\gamma 1\alpha_{MM}$ or BK $\gamma 1\alpha_{M(dual)}$. To express the BK $\gamma 1\alpha_{MMMM}$ (also referred as BK $\gamma 1\alpha_{M(quad)}$) fusion construct, we created pcDNA6-BK $\gamma 1\alpha_{M1}\alpha_{M2}\alpha_{M3}\alpha_{M4}$ -V5-His by subcloning $\alpha_{M3}\alpha_{M4}$ from pcDNA6- $\alpha_{M3}\alpha_{M4}$ -V5-His into pcDNA6-BK $\gamma 1\alpha_{M1}\alpha_{M2}$ -V5-His using NotI and AgeI. For DNA manipulation and amplification, we used the Long Fragment DNA Ligation Kit (TaKaRa) and CopyCutter competent *E. coli* (Lucigen). Site-directed mutagenesis was performed using the QuickChange kit (Stratagene). Mutations for the quadruple-repeat construct were first introduced into either the $\alpha_{M1}\alpha_{M2}$ or $\alpha_{M3}\alpha_{M4}$ dual-repeat construct, followed by fusion of $\alpha_{M1}\alpha_{M2}$ and $\alpha_{M3}\alpha_{M4}$ as described above. HEK293 cells were cultured in DMEM supplemented with 10% fetal bovine serum at 5% CO₂. Cells were transfected with plasmids using PEI "MAX" (Polysciences Inc.) and subjected to electrophysiological assays 16-72 hours post-transfection.

Electrophysiology

BK channel currents were recorded from excised inside-out patches of HEK293 cells using patch-clamp recording techniques as described previously (60). Both intracellular and extracellular (pipette) solutions contained 136 mM KMeSO₃, 4 mM KCl, and 20 mM HEPES (pH 7.20). The extracellular solution was supplemented with 2 mM MgCl₂, while the intracellular solution contained 5 mM HEDTA either with or without Ca²⁺ to achieve 10 μ M Ca²⁺ or Ca²⁺-free. Recording pipette electrodes were pulled from borosilicate filamented glass tubes (Cat #: BF150-110-10, Sutter Instrument) with a P-1000 micropipette puller (Sutter Instrument), and polished by heat with an MF-830 microforge (Narishige) to a resistance of 1-2 M Ω . Data were acquired using PatchMaster (HEKA) with an Axopatch 200B amplifier (Molecular Devices) and ITC-18 digitizer (InstruTECH) or with an EPC-10 amplifier (HEKA). Data were sampled at 20 μ s and filtered at 2 kHz (Axopatch 200B) with the amplifiers' 4-pole Bessel filter or at 2.9 kHz (EPC-10). Time interval between voltage protocol sweeps was 2 s. Capacitive and leak currents were subtracted using a P/4 protocol at holding potentials of -120 mV or -150 mV (for $\gamma 1$ or 10 μ M Ca²⁺ conditions). Steady-state activation, expressed as normalized conductance (G/G_{max}) versus voltage (G-V), was calculated from the tail current amplitudes (at -120 mV) and fitted using a single-Boltzmann function $G/G_{max} = 1/(1 + e^{-ZF(V-V_H)/RT})$ or a double-Boltzmann function $G/G_{max} = P_a/(1 + e^{-ZaF(V-V_{H1})/RT}) + (1 - P_a)/(1 + e^{-ZbF(V-V_{H2})/RT})$ where V, V_H, Z, F, R, T, P_a, a and b denote voltage, V_{1/2}, gating charge (z), Faraday constant, gas constant, Kelvin temperature, component portion (0-1), and component identity (a or b), respectively. Values are reported as means $\pm$ SEM.

Immunoblotting

Proteins were enriched by immunoprecipitation as previously described (19) and immunoblotted after SDS-PAGE. Briefly, proteins were solubilized from cells in 2% Dodecyl-beta-D-maltoside (DDM) in TBS buffer (50 mM Tris, 150 mM NaCl, pH 7.6). Lysates were incubated with mouse anti-V5 monoclonal antibody agarose gel (Cat# A7345, Millipore Sigma) at 4 $^{\circ}$ C for 2 h. After three 10 min washes in with 2% DDM-containing TBS, bound proteins were eluted with 4% SDS. Protease

inhibitor cocktail (Roche) was used throughout the procedure. Eluted proteins were separated by 4% to 20% gradient SDS-PAGE and transferred to PVDF membranes. Immunoblotting was performed with mouse anti-V5 monoclonal antibody (Cat# R96125, Invitrogen) at 1:10,000 dilution.

Figure supplement

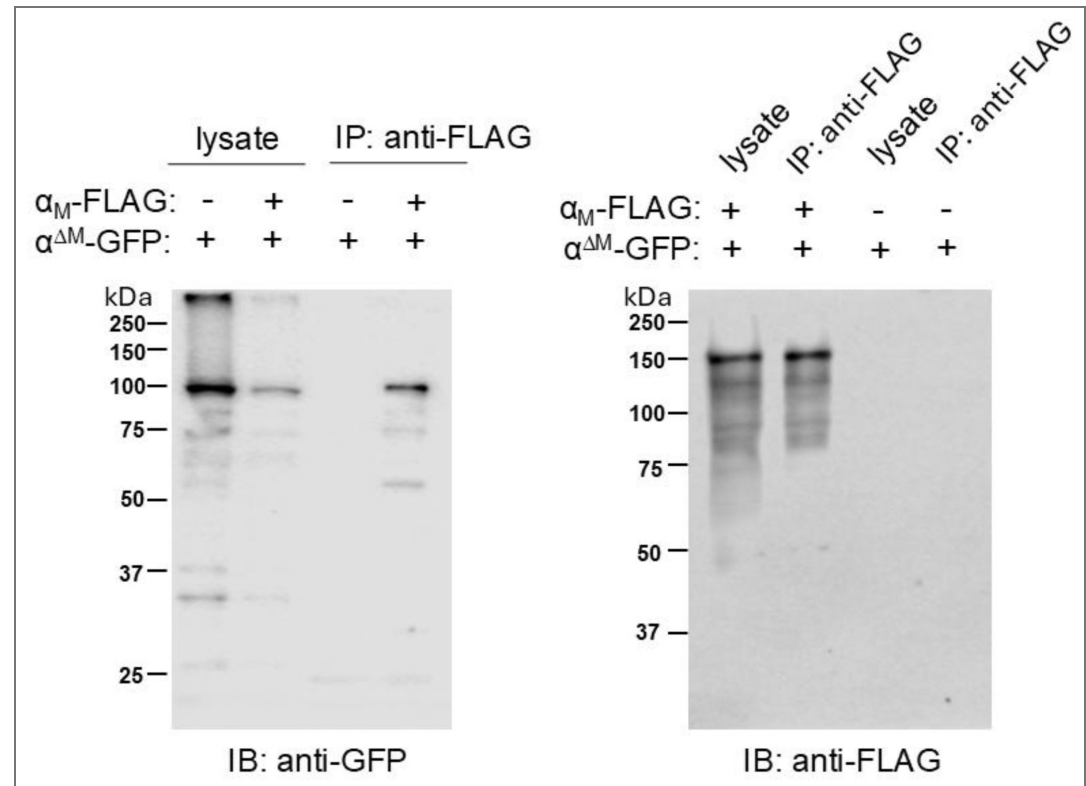


Figure 1 - figure supplement 1. Complex formation between split BKα subunit constructs. A FLAG-tagged α_M construct was co-expressed with GFP-tagged $\alpha_{\Delta M}$ or an empty vector in HEK293 cells. For comparison, equal amounts of total cellular protein from cell lysates were loaded onto SDS-PAGE. Notably, the presence of α_M caused a reduction of the $\alpha_{\Delta M}$ expression and the singular α_M construct mostly oligomerized on SDS-PAGE.

Data availability

All data relevant to this work is presented in the manuscript and supporting files.

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Additional information

Author Contribution

G.C., Q.L., and J.Y. designed experiments and analyzed data. G.C., Q.L., and K.S. performed experiments. G.C. and J.Y. wrote the manuscript.

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Additional files

[Figure 1 - Source data](#)

[Figure 2 - Source data](#)

[Figure 3 - Source data](#)

[Figure 4 - Source data](#)

[Figure 1C - Source data](#)

[Figure 2C - Source data](#)

[Figure 1 - figure supplement - Source data](#)

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Peer reviews

Reviewer #1 (Public review):

Summary:

BK channels are widely distributed and involved in many physiological functions. They have also proven a highly useful tool for studying general allosteric mechanisms for gating and modulation by auxiliary subunits. Tetrameric BK channels are assembled from four separate alpha subunits which would be identical for homozygous alleles and of potentially five different combinations for heterozygous alleles Geng et al . (2023), (<https://doi.org/10.1085/jgp.202213302>). Construction of BK channels with concatenated subunits in order to strictly control heteromeric subunit composition had not yet been used because the N-terminus in BK channels is extracellular whereas the C-terminus is intracellular. In this new work, Chen, Li, and Yan devise clever methods to construct and assemble BK channels of known subunit composition, as well as to fix the number of γ1 axillary subunits per channel. With their novel molecular approaches, Chen, Li and Yan report that a single γ1 axillary subunit is sufficient to fully modulate a BK channel, that the deep conducting pore mutation L312A exhibited a graded effect on gating with each addition mutated subunit replacing a WT subunit in the channel adding an additional incremental left shift in activation, and that the V288A mutation at the selectivity filter must be present on all four alpha subunits in order to induce channel inactivation. Chen, Li, and Yan have been successful in introducing new molecular tools to generate BK channels of known stoichiometry and subunit composition. They validate their methods and provide three different examples of stoichiometric modulation by LRRC26, the selectivity filter, and the pore.

Strengths:

Powerful new molecular tools for study of channel gating are developed and validated in the study.

Weaknesses:

One example each of auxiliary, deep pore, and selectivity filter allosteric actions are presented, but this is sufficient for the purposes of the paper to establish their methods and present specific examples of applicability.

<https://doi.org/10.7554/eLife.107681.2.sa2>

Reviewer #2 (Public review):

Summary:

This manuscript describes novel BK channel concatemers as a tool to study the stoichiometry of gamma subunit and mutations in modulation of the channel. Taking the advantage of modular design of BK channel alpha subunit the authors connected S1-S6/1st RCK as two- and four-subunit concatemers and coexpressed with S0-RCK2 to form normal function channels. These concatemers avoided the difficulty that the extracellular N-terminus of S0 was unable to connect with the cytosolic C-terminus of the alpha or gamma subunit, allowing a single gamma subunit to be connected to the concatemers. The concatemers also helped reveal the required stoichiometry of mutant BK subunits in modulating channel function. These include L312A in the deep pore region that altered channel function additively with each additional subunit harboring the mutation, and V288A at the selectivity filter that altered channel function cooperatively only when all four subunits being mutated. These results demonstrate that the concatemers are robust and effective in studying BK channel function and molecular mechanisms related to stoichiometry. The different requirement of the gamma subunit and the mutations stoichiometry for altering channel function is interesting, revealing fundamental mechanisms of how different motifs of the channel protein control function.

Strengths:

The manuscript presents well designed experiments with high quality data, which convincingly demonstrate the BK channel concatemers and their utility. The results are clearly written.

Weaknesses:

This reviewer did not identify any major concerns with the manuscript.

Editors' note: We thank you for addressing some of the concerns, adding clarifications and more complete discussions, including further details about experimental protocols. The revised version is significantly improved. Some concerns linger that the biophysical/structural mechanisms underlying the observed phenotypes remain unclear and in some ways are phenomenological. However, the current study is more about the methodology and the mechanisms underlying the stoichiometry dependent effects are perhaps left for a separate study, with more detailed exploration. Congratulations for the excellent work.

<https://doi.org/10.7554/eLife.107681.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary:

BK channels are widely distributed and involved in many physiological functions. They have also proven a highly useful tool for studying general allosteric mechanisms for gating and modulation by auxiliary subunits. Tetrameric BK channels are assembled from four separate alpha subunits, which would be identical for homozygous alleles and potentially of five different combinations for heterozygous alleles (Geng et al., 2023, <https://doi.org/10.1085/jgp.202213302>). Construction of BK channels with concatenated subunits in order to strictly control heteromeric subunit composition had not yet been used because the N-terminus in BK channels is extracellular, whereas the C-terminus is intracellular. In this new work, Chen, Li, and Yan devise clever methods to construct and assemble BK channels of known subunit composition, as well as to fix the number of $\gamma 1$ axillary subunits per channel. With their novel molecular approaches, Chen, Li and Yan report that a single $\gamma 1$ axillary subunit is sufficient to fully modulate a BK channel, that the deep conducting pore mutation L312A exhibited a graded effect on gating with each addition mutated subunit replacing a WT subunit in the channel adding an additional incremental left shift in activation, and that the V288A mutation at the selectivity filter must be present on all four alpha subunits in order to induce channel inactivation. Chen, Li, and Yan have been successful in introducing new molecular tools to generate BK channels of known stoichiometry and subunit composition. They validate their methods and provide three examples of their use with useful observations.

Strengths:

Powerful new molecular tools for the study of channel gating have been developed and validated in the study.

Weaknesses:

(1) One example each of auxiliary, deep pore, and selectivity filter allosteric actions is presented, but this is sufficient for the purposes of the paper to establish their methods and present specific examples of applicability.

We sincerely thank Reviewer #1 for the thoughtful and supportive evaluation of our work. We greatly appreciate the reviewer's clear summary of the study and the recognition of the novelty and utility of our molecular concatemer strategy for controlling BK channel subunit composition and stoichiometry.

We also appreciate the reviewer's positive assessment that the three examples (auxiliary subunit modulation, deep pore mutation, and selectivity filter mutation) are sufficient to establish the method and demonstrate its applicability. We are encouraged that the reviewer found the new molecular tools to be powerful and well validated.

We have no further changes to make in response to this review, but we are grateful for the reviewer's constructive and encouraging comments.

Reviewer #2 (Public review):

Summary:

This manuscript describes novel BK channel concatemers as a tool to study the stoichiometry of the gamma subunit and mutations in the modulation of the channel. Taking advantage of the modular design of the BK channel alpha subunit, the authors connected S1-S6/1st RCK as two- and four-subunit concatemers and coexpressed with S0-RCK2 to form normal function channels. These concatemers avoided the difficulty that the extracellular N-terminus of S0 was unable to connect with the cytosolic C-terminus of the gamma subunit, allowing a single gamma subunit to be connected to the concatemers. The concatemers also helped reveal the required stoichiometry of mutant BK subunits in modulating channel function. These include L312A in the deep pore region that altered channel function additively with each additional subunit harboring the mutation, and V288A at the selectivity filter that altered channel function cooperatively only when all four subunits were mutated. These results demonstrate that the concatemers are robust and effective in studying BK channel function and molecular mechanisms related to stoichiometry. The different requirement of the gamma subunit and the mutations stoichiometry for altering channel function is interesting, which may relate to the fundamental mechanism of how different motifs of the channel protein control function.

Strengths:

The manuscript presents well-designed experiments with high-quality data, which convincingly demonstrate the BK channel concatemers and their utility. The results are clearly presented.

Weaknesses:

This reviewer did not identify any major concerns with the manuscript.

We sincerely thank Reviewer #2 for the careful reading of our manuscript and for the highly positive and supportive comments. We appreciate the reviewer's detailed summary of our concatemer design strategy and its use in studying gamma subunit stoichiometry and mutation-dependent modulation of BK channel function.

We are especially grateful for the reviewer's recognition that the experiments are well designed, the data are of high quality, and the results demonstrate the robustness and utility of the concatemer approach. We also appreciate the reviewer's thoughtful note on the mechanistic implications of the distinct stoichiometric requirements observed for the gamma subunit, L312A, and V288A.

We are pleased that the reviewer identified no major concerns. We have no further changes to make in response to this review, and we thank the reviewer again for the positive evaluation.

Recommendations for the authors:

Reviewing Editor Comments:

While the study presents a great methodological advancement, the phenomenological examples described could perhaps benefit from a little more mechanistic description/discussion. In particular, the functional effect of the V288A mutant is very novel. It could be useful to discuss whether this mutant impacts channel selectivity/conductance. It could be beneficial to also contrast the subunit dependence of V288A with that of the W434F mutant of the Shaker channel. In the latter, C-type inactivation gating is accelerated even when the mutant is present in a single subunit, which contrasts with the effect in V288A.

We greatly appreciate the editor's and reviewers' thorough and constructive evaluation, and we have revised the manuscript accordingly.

We added discussion with citation about the potential effect of V288A on selectivity (lines 348349). We also added the reported stoichiometric effects of mutations in Shaker and hERG1 channels on C-inactivation in discussion (lines 336-351). From these studies and our findings with V288A in BK channels, it is interesting to note that the stoichiometric effects of these mutations varies and those located near or within selectivity filter signature exhibited an all-or-none effect in both hERG1 and BK channels.

The authors might also want to consider performing and showing immunoblots with the alpha_deltaM fragment co-expressed with the other channel fragments. Together with the GFP tag, this alpha_deltaM would perhaps be a ~90 kDa protein. It should be captured by anti-V5 IP and resolved on an SDS-PAGE gel (at least with the quad construct).

We added supplemental data (Fig.1 – figure supplement 1) to show co-expression and co-IP of the $\alpha^{\Delta M}$ -GFP construct and a FLAG-tagged α_M construct. The $\alpha^{\Delta M}$ -GFP displayed right size on SDS-PAGE. It is of note that the single unit α_M construct tended to oligomerize even under denatured condition on SDS-PAGE.

For Figure 4, providing details about the inter-pulse intervals and interpulse holding voltage would be helpful. I was not able to find this information in the methods or text.

The inter-pulse intervals and holder voltage are now added in Fig. 4 legend (line 638).

Reviewer #1 (Recommendations for the authors):

(1) Submitted papers should have page numbers to facilitate reviewing.

Both page and line numbers are added.

(2) The designation of the various channel types, such as BK α and BK α_M should be identical in the text and figures, so either drop BK in the text or add BK in the figures. Maybe drop BK in the text, as it is known that BK channels are the topic of this study.

We appreciate the suggestion to be consistent in text and figures. We have dropped “BK” for “BK α_M ” throughout the text.

(3) "Single Boltzmann fits of G-V curves" would be consistent with a homogenous channel population but do not necessarily suggest a single homogenous channel population of BK channels, as was shown by Geng et al. (2023) (<https://doi.org/10.1085/jgp.202213302>) where the G-V curve for simultaneous expression of five BK channel types with different V1/2s for each channel type was well approximated by a single Boltzmann function. The dogma that a single Boltzmann fit suggests one channel type needs to be reset. So wave a red flag here: whereas a single Boltzmann fit is consistent with a single channel type, it does not establish a single channel type nor even suggest a single channel type.

We fully agree that a good Single Boltzmann fit doesn't mean homogenous channel population. We have changed “suggesting” to “consistent with” (line 203) and “reflecting” to “agreeing with” (line 205).

(4) Geng et al. (2023) demonstrated that the pore mutation G375R in BK channels gave a left shift in activation linearly related to the number of WT subunits replaced with mutant subunits. This should incremental shift in activation for G375R should be mentioned, as it

is consistent with the incremental effects of the L312A deep pore mutation on activation as reported by the authors in their Figure 3D.

We appreciate the pointing-out of this highly relevant publication. We have now included this reference and discussed together with L312A mutation (lines 309-313).

(5) I went back and looked at the Lingle laboratory papers on the gamma subunit. An additional sentence or two on what the Lingle lab found and didn't find would be useful here for readers.

In the Introduction, we have listed the Lingle lab's findings and the limitations of their experimental methods that warrants the development of a concatenated construct method as proposed in this study (lines 84-88). We prefer to not discuss further in the Discussion as it will be redundant.

(6) For the two examined mutations L312A and V288A, include in the Methods a 21 amino acid sequence for each mutation with the amino acid to be mutated (L or V) in the center, with beginning and end numbering at the beginning and end of each list. This will allow the reader/experimenter to readily locate the mutated residue on their BK amino acid sequences, which may have different numbering than U11058. Interestingly, for the so-called canonical sequence Q12791 · KCMA1_HUMAN that I found in UniProt starting with U11058, there is an L312, but I found no V288, but an F288. Am I doing this correctly? Do I have the correct sequence/isoform? The only sure way to identify an AA is with an extensive pre and post-sequence so that the chance of misidentification approaches zero.

We verified that the listed Gene Bank IDs of U11058 for cDNA and AAB65837 for protein should point to the right sequences. In the section of Results, we have now included the peptide sequences of the selectivity filter signature motif and part of the S6 TM where V288 and L312A are located, respectively (lines 179 and 220).

Reviewer #2 (Recommendations for the authors):

The different stoichiometry of the gamma subunit and the mutations in regulating channel function raise important questions. For instance, what are the structural and energetic bases for their different stoichiometric requirements? Does the structure motif, such as the selectivity filter or deep pore, act as a unit? Or does a specific residue, such as V288 or L312, act individually to determine the different stoichiometric requirements? What molecular interactions are involved for these residues and subunit to influence the cooperativity among the four alpha subunits in channel function? Some of these questions are discussed in the manuscript, but it may help the readers to clarify what aspects of the mechanistic bases for the findings in this manuscript are known and what aspects remain to be studied.

We agree that these are all important questions. We have now cited more previous studies on C-inactivation in other K⁺ channels and on deep pore mutations in BK channels in terms of subunit stoichiometry (lines 336-351). The results appear to be consistent, suggesting shared properties among residues within the selectivity filter motif or among residues in deep pore region.

Some minor comments are as follows.

(1) Page 7, 2nd paragraph: "Page 2B" change to "Page 3B"? Also, "delay in deactivation" is not precise. The term "Delay" in channel kinetics has a specific meaning, and the use of this word here causes some confusion. The authors may want to delete "substantial delay in deactivation evident as a".

Corrected by changing Fig. 2B to Fig. 3B and deleting “a substantial delay in deactivation evident as” (line 191).

(2) Page 9, 1st paragraph: “used in the voltage protocol used”. Drop one of the instances of used”.

Corrected by deleting the first “used” (line 246).

(3) Page 12, 1st paragraph: “Nonetheless, the tight inter-subunit cooperativity observed at the selectivity filter makes it a plausible candidate for serving as the activation gate, a property not yet demonstrated for the lower S6 segment.” This seems to be an interesting idea. However, it is not clearly explained. The authors may want to clarify how the cooperativity is related to the activation gate.

We have now added a sentence with citations to discuss the requirement of intersubunit cooperativity for an activation gate to function (lines 354-357).

Other major changes: We updated immunoblot figures Fig1C and Fig2C for better presentation.

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